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BMP-4 Extraction from Extracellular Matrix and Analysis of Heparin-Binding Properties

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Abstract

Recombinant human BMP-4 growth factor (GF) has significant commercial potential as therapeutic for regenerating bone and as cell culture supplement. However, its commercial utility has been limited as large-scale attempts to express and purify human BMP-4 GF have proved challenging. We have established a novel approach to obtain significant quantities of pure and bioactive BMP-4 GF from Chinese hamster ovary cell cultures by extracting the GF moiety from the extracellular matrix or cell pellet fraction. This approach increased yields approximately one 100-fold over BMP-4 GF purified from CM. The molecular activities of the two fractions are indistinguishable. We further analyzed binding of BMP-4 GF to the proteoglycan Heparin and showed that an N-terminal basic sequence is essential for this interaction. Taken together, these results provide novel insights into the purification, localization, and Heparin binding of human BMP-4 that have implications for its bioprocessing and biological function.

Keywords

Bone morphogenetic protein; BMP-4; TGF- β ; Purification; CHO; Mammalian cell culture; Extracellular matrix; ECM; Heparin

Introduction

Bone Morphogenetic Protein 4 (BMP-4) is a member of the Transforming Growth Factor- β (TGF- β) family that was originally identified from bone extracts where it co-purified with its close homolog BMP-2 [1]. BMP-4 has therapeutic potential for treating long bone fractures, osteoarthritis, rheumatoid arthritis, and for regenerating cartilage [2]. In addition, BMP-4 is an essential regulator of developmental processes, including mesoderm formation and neural differentiation [3]. In vitro, BMP-4 can sustain self-renewal of naive mouse Embryonic Stem

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Conflict of interest EMH holds shares of Acceleron Pharma. EMH is a shareholder and co-founder of Advertent Biotherapeutics. SA is an employee and holds stock options of Regeneron Pharma. JM has no competing interests.

Ethical Approval This article does not contain any studies with human participants or animals.

Informed Consent All the authors gave consent for publication.

cells (ESCs) together with Leukemia Inhibitory Factor (LIF) and is used as substitute for serum in human ES/iPS cell expansion media [4, 5]. Based on its many desirable biological properties, recombinant BMP-4 has significant commercial potential as therapeutic, for the maintenance of mESC cultures and for the differentiation of embryonic and induced pluripotent stem cells into various lineages.

Although many recombinant TGF- β family growth factors (GFs) have been produced, large-scale attempts to express and purify human BMP-4 GF have proved especially challenging. One general reason for this difficulty is the complex biological synthesis of all TGF- β family GFs. TGF- β family GFs are expressed as precursors that consist of signal peptide, propeptide, and growth factor (GF) moiety. Furin proteases cleave precursors at an RXXR sequence during secretion to release the mature, dimeric GF [6, 7]. The highly disulfide-linked GF structure poses an additional protein-folding challenge for making bioactive TGF- β family GFs [8]. As a result, mammalian expression systems are the preferred method for expressing TGF- β family GFs. Nevertheless, efficient production of BMP-4 remains notoriously difficult. Thus, several reports describe complex approaches to improve yields. These approaches involve swapping of pro-domains, genomic alterations of the expression cell line, and multi-site mutagenesis [9–16]. Refolding from *E. coli* has also been reported [17, 18]. However, most refolded forms lack 10 N-terminal residues of the GF domain and their activity is not always quantitatively defined. Notably, the deleted 10 N-terminal residues are thought to be required for the interaction between BMP-4 and extracellular matrix (ECM) proteoglycans [19, 20]. The N-terminal sequence may, therefore, be fundamentally important for the biological action of BMP-4. Moreover, BMP-4 produced in *E. coli* is not glycosylated. These caveats raise the question of functional equivalence between the mammalian cell culture made BMP-4 GF and refolded forms.

To address the problem of yield and activity of the naturally occurring and glycosylated BMP-4 GF overexpressed in mammalian cells, we developed a novel approach for its extraction from the cell pellet fraction/extracellular matrix of stably transfected Chinese hamster ovary cells. We compared biological activities of BMP-4 GF produced using this approach with commercial preparations made using mammalian cells and with the refolded form that lacks 10 N-terminal amino acids. Our results show that our extraction protocol can significantly improve the yield of biologically active BMP-4 GF. In addition, our comparisons reveal functional differences between mammalian cell produced and commercially available, refolded BMP-4.

Construct Design and Expression

The sequence alignment in Fig. 1A reveals key similarities and differences between human BMP-4, the refolded form that lacks 10 N-terminal residues and its close homolog BMP-2. The majority of the GF region is highly conserved between BMP-2 and BMP-4. However, the basic N-terminal region is highly divergent between the two growth factors. The N-terminal sequence is truncated in one commercially available, refolded form (BMP-4 Δ).

To express BMP-4 in Chinese hamster ovary cells (CHO), we created 3 constructs that included a non-native signal peptide and a His₈ tag followed either by Maltose-Binding

Protein (MBP) and human BMP-4 (MB4), human BMP-4 (B4/4), or a chimera of the BMP-2 pro-domain and the BMP-4 mature domain (B2/4) (Fig. 1B). Constructs were co-expressed with or without Furin. Cells were transfected and subjected to various levels of puromycin/methotrexate selection. Briefly, in a first round of selection cells were treated with 20 µg/mL puromycin and 200 nM methotrexate. In a second round of selection cells were treated with 30 µg/mL puromycin and 500 nM methotrexate. To establish the productivity of the various constructs and selections, we screened total BMP-4 expression using Western blot (Fig. 1C). Analyzed samples consisted of equivalent volumes of cell suspensions with a defined number of cells, *i.e.*, samples included conditioned medium and cells. Overall, all but one construct expressed BMP-4 growth factor (GF, BMP-4^{GF}), as evidenced by the 12–15 kDa band that reacted with the anti-BMP-4 antibody. The exception was MBP-BMP-4 co-expressed with furin, where neither GF nor unprocessed (UP, BMP-4^{UP}) forms were detected. Cell productivity declined with more stringent selection, as indicated by the reduced levels of BMP-4 in samples subjected to two rounds of selection. Furin co-expression improved processing, as suggested by the loss of the unprocessed forms (UP, MUP). However, it did not lead to increased yield, as indicated by similar or reduced levels of GF.

BMP-4 GF Associates with the Cell Pellet Fraction

Like all TGF-β family GFs, BMP-4 is secreted. We, therefore, expected to find the majority of the processed GF in the conditioned medium (CM) of our CHO cell cultures. However, we discovered that most cultures did not have appreciable levels of BMP-4 in the CM (Fig. 2A). To establish the location of the BMP-4 GF that was detected in the total sample, we analyzed various fractions, including CM, cell pellet, and cell pellet wash fractions (Fig. 2A). Fractionation indicated that the GF moiety preferentially associated with the cell pellet. One exception was the chimeric B2/4 construct, which also had significant levels of GF in the CM. This observation is consistent with a previous report showing that B2/4 gives far better yields of mature BMP-4 GF in CM than wild type BMP-4 (B4/4) [9]. For this reason, B2/4 is likely the main source of BMP-4 GF produced in mammalian cells to date. To determine if BMP-4 GF can be released from the cell pellet fraction (CPF), we washed cells in an equivalent volume of Tris-buffered saline (TBS) and collected the supernatant after centrifugation. We detected BMP-4 GF in the TBS supernatant of several constructs, indicating that BMP-4 GF can be extracted from the CPF using a wash step. Notably, B2/4 expressed without furin yielded the highest levels of mature BMP-4 GF from the CPF. To determine if the pro-domain remained associated with the GF, we probed its localization using an anti-His antibody. We found that the pro-domain was present both in the CM and CPF (Fig. 2B), indicating that a significant pro-domain fraction is dissociated from the GF moiety.

Dimeric BMP-4 GF can be Eluted from the Cell Pellet Fraction

To identify optimal conditions for BMP-4 extraction from the CPF, we treated samples of wild type BMP-4 (B4/4) that was co-expressed with furin to a variety of buffers and sonication (Fig. 3A). We speculated that high salt buffers would elute the GF from the membrane as BMP-4 and other members of the family are known to associate with

glycosa-minoglycans (GAGs) on the extracellular matrix (ECM) [21, 22]. Although Heparin treatment released BMP-4 from the cell pellet, high ionic strength buffers did not. In fact, they blocked GF release by Heparin. Strikingly, low ionic strength solutions including ddH₂O and buffer conditions used to elute BMP-4 from hydrophobic or Heparin affinity chromatography media effectively extracted the GF moiety from the CPF, as indicated by the strong signal obtained with an anti-BMP-4 antibody immune blot. These results, therefore, show that substantial levels of BMP-4 GF can be extracted from the CPF using a combination of low ionic strength buffers and sonication. Notably, low ionic strength elution conditions suggest that the GF moiety may interact with proteoglycans and other, as yet unidentified components of the ECM.

To evaluate the receptor-binding activity of the extracted samples, we injected equivalent volumes of the different supernatants onto an SPR sensor chip containing Fc-fusions of the BMP-4 receptor ALK3 (Fig. 3B) [23, 24]. We found that BMP-4 GF released by the low ionic strength buffer containing 0.5 M Arginine and 20% propylene glycol (APG) had significant receptor-binding activity as indicated by the SPR response (Fig. 3B, blue curve). To compare activities of the different BMP-4 constructs, we treated the CPF with APG buffer and injected equivalent volumes of supernatant onto an SPR sensorchip containing either ALK3-Fc or ActRIIA-Fc (Fig. 3C). We found that GF-binding activity could be detected after extraction from all constructs and that the binding properties of the different samples were comparable, as indicated by the indistinguishable kinetic profiles. However, SPR responses were different as extracted GF concentrations varied. By contrast, CM samples did not show a significant SPR response, reflecting either lower levels of BMP-4 in CM, inactive GF in the CM, or a general difficulty in evaluating the activity of BMP-4 GF directly from CM. Thus, these results demonstrate that significant amounts of active BMP-4 GF can be obtained by releasing the GF moiety from the extracellular matrix using low ionic strength buffers such as APG.

BMP-4 GF can be Purified to Homogeneity from the Cell Pellet Fraction

To purify the various BMP-4 forms and fractions, we developed three purification strategies that take into account the properties of each sample. These strategies are based on published reports or patents (Fig. 4A) and are described in detail in the methods section [21, 25].

We first attempted to purify BMP-4 GF from CM based on a combination of previously published protocols that included a Heparin capture step, hydrophobic interaction, and ion exchange chromatography (Fig. 4B). A final, reversed phase chromatography step was added to improve purity and to obtain GF in a buffer that can be easily removed using a Speed-Vac concentrator. We found this step to be critical as dialysis resulted in drastically reduced yields, complicating any buffer exchange steps. As expected based on our previous observations, only the B2/4 construct yielded acceptable levels of GF from the CM (Table 1). The BMP-4 GF moiety and BMP-2 pro-domain were separated during the Heparin column binding step, where the pro-domain was in the flow-through (FT) and the GF moiety bound to the column. We further purified the BMP-4 GF moiety using a Butyl Sepharose hydrophobic interaction chromatography step, where BMP-4 was eluted using the APG buffer. This fraction was then loaded on sequentially connected Anion and Cation exchange

columns (HiTrap Q and SP, respectively). BMP-4 GF flowed through the Anion Exchange column and was captured by the Cation Exchange column. As a final purification step, the HiTrap SP-eluted GF moiety was subjected to Reversed-Phase Chromatography (RPC). In contrast to the CM fraction, ECM-associated BMP-4 was captured from the supernatant fraction of the cell pellet (CPF). Briefly, we subjected the cell pellet to sonication in APG buffer (which is used to elute BMP-4 from Butyl Sepharose hydrophobic interaction media). We then adjusted the conductivity of this sample and, as with CM-purified BMP-4, captured the BMP-4 GF moiety using Butyl Sepharose hydrophobic interaction media (Fig. 4C). Residual pro-domain was separated during the Butyl Sepharose wash step and the GF moiety was purified to homogeneity using the same procedure developed for BMP-4 GF captured from CM. To obtain unprocessed BMP-4, we purified MBP-BMP-4 (MB4) from CM. This construct was extremely sensitive to degradation and we could only isolate low levels of unprocessed BMP-4 following capture by immobilized metal affinity chromatography.

To gain insights on yields, we compared equivalent volumes of RPC-purified BMP-4 GF with commercial BMP-4 (RnD, Fig. 4D). The top Western blot shows the different BMP-4 fractions, while the bottom blot shows various loading amounts of commercial BMP-4. These results show that the levels of CM-purified BMP-4 GF are low, except for the B2/4 construct expressed without furin (lane 4). Levels of BMP-4 GF extracted from the CPF are high, indicating that this is a far better source of mature GF. Overall, the B2/4 construct performed like wildtype, except for the CM fraction, where the B2/4 construct gave significantly higher yields of GF. By contrast, the MBP fusion approach resulted in lower levels of mature GF and most was found unprocessed in CM. Comparison of the upper and lower Western blots and Bicinchoninic Acid (BCA) quantitation indicates that our high yielding purifications (lanes 4–8) produce 1–2 mg BMP-4 GF per liter culture (Table 1).

To ascertain that the purified GF forms disulfide-linked dimers, we compared the electrophoretic mobility of reduced and non-reduced samples (Fig. 4E). The Coomassie-stained gel shows that all purified GF are dimeric (samples 1–4), as indicated by the molecular weight of about 35 kDa and 18 kDa for the non-reduced and reduced samples, respectively. Similarly, the silver-stained gel (sample 5) shows that full-length, unprocessed BMP-4 purified as described above is also dimeric, indicating that dimerization occurs before proteolytic processing. We note that the GF purified from the B2/4 construct expressed without furin exhibits more diffuse bands (lanes 8/9), possibly indicating that post-translational processing may be more homogeneous when BMP-4 is expressed with its native pro-domain and furin. In conclusion, these data show that the best yields of BMP-4 GF are from the cell pellet fraction and we find that the B2/4 chimera improves yields over native B4/4 from both CM and CPF.

BMP-4 Purified from the Cell Pellet Fraction is Active

Mature BMP-4 GF exerts its function by binding two type I and two type II transmembrane TGF- β family receptors to activate an intracellular signaling cascade that involves SMAD transcription factors [26]. Specifically, BMP-4 interacts with the type I receptors ALK3/ALK6 and the type II receptors ActRIIA/ActRIIB and BMPRII. To compare the activities

of the different BMP-4 preparations, we investigated their binding to ALK3 and ActRIIA using SPR. We picked these two receptors as proxies, as we had limited access to the other receptors. We analyzed four samples, B2/4 minus furin purified from CM, B2/4 minus furin purified from the CPF, B4/4 minus furin purified from the CPF, and unprocessed MBP-BMP-4 purified from CM. Using the same concentration of BMP-4 and single injections, we found that all RPC-purified BMP-4 GF preparations bind their type I receptor ALK3 and type II receptor ActRIIA with equivalent potency, as indicated by the nearly identical binding amplitude and kinetic profile (Fig. 5A). Although the binding amplitude of the B2/4 CPF to ActRIIA was larger, the profile was similar, suggesting that this sample may have a non-specific or a bulk contribution. Notably full-length, unprocessed BMP-4 did not bind its cognate receptors, suggesting that this form is latent (black line). To show that our different constructs produced BMP-4 GF of similar activity, we carried out an SPR titration of the B4/4 and B2/4 GF moieties, where a range of GF concentrations were used. We found that both bind the receptor ALK3 with indistinguishable binding rates (Fig. 5B).

We then compared our B4/4 GF samples with commercially available growth factors, including BMP-2, refolded BMP-4 Δ , and cell culture produced, commercial BMP-4 (BMP-4^C). We noticed several surprising differences (Fig. 5C). Namely, all samples bind the type I receptor ALK3 with nearly identical kinetics and affinity, but interactions with the type II receptor ActRIIA differed significantly. Thus, in contrast to our BMP-4 GF preparations, BMP-2 and BMP-4 Δ have fast on and off rates, as evidenced by the sharp increase of the SPR response during the association phase and a sharp decrease during the dissociation phase (Table 2). We note that BMP-4 Δ differs from our preparations in that it lacks 10 N-terminal amino acids, is refolded from *E. coli* inclusion bodies, and is not glycosylated, while BMP-2 is almost identical to BMP-4 except for its 10 amino-terminal residues (Fig. 1A). In sum, these results show that mammalian made BMP-4 GF binds type II receptors with a kinetic profile that is distinct from BMP-2 and refolded BMP-4 Δ . Although we found that BMP-4^C had a smaller ActRIIA binding amplitude, we speculate that this form may have lost activity due to storage or formulation, a problem commonly encountered with this GF.

To compare the biological activity of the different forms and constructs, we evaluated their signaling potency using a SMAD1/5/8 responsive reporter gene expression assay [27]. Most preparations had comparable potency at 10 nM concentration as indicated by similar SMAD1/5/8-dependent BRE reporter signal amplitudes (Fig. 5D). Exceptions were full-length MBP-BMP-4, which may be latent, and CM fractions, which did not yield adequate levels of purified GF. We note that only CM from the BMP2/4 construct expressed without Furin yielded sufficient GF for this analysis. Dose-response curves indicated that BMP-4 and BMP-2 reached half-maximal response at similar concentrations (Fig. 5E, Table 2), including our B4/4 GF sample and BMP-4^C. However, BMP-4 Δ was less potent as indicated by an approximately tenfold lower EC₅₀ value.

BMP-4 Binding with Heparin Blocks Type I Receptor Interaction

The bioavailability of many TGF- β family ligands is regulated by their association with the extracellular matrix (ECM) [22]. Indeed, ECM proteoglycans, such as Heparin, may bind

basic amino-terminal sequences of BMP-4 and BMP-2 to restrict their diffusion in vivo [19, 20]. Notably, BMP-4 Δ lacks these amino-terminal residues and BMP-2 and BMP-4 differ most significantly in this region (Fig. 1A).

To determine how proteoglycans regulate the activities of these GFs, we investigated their interaction with Heparin (Fig. 6). Using SPR, we first determined how the different GFs bind Heparin directly (Fig. 6A, Table 2). We captured biotinylated Heparin on a streptavidin-coated surface for binding analysis [28]. Using single injections at one ligand concentration (25 nM), we found that BMP-2 and CPF/ECM extracted BMP-4 (B4/4-GF) bound Heparin with high affinity, while the affinity of BMP-4 Δ for Heparin, calculated based on a two-state reaction model, was approximately one order of magnitude lower and exhibits significant differences both in the association and dissociation rates (Fig. 6B, Table 2).

To elucidate the molecular role of Heparin in the biochemistry of the different GFs, we evaluated its effect on ligand-receptor binding. Briefly, we incubated ligands with increasing Heparin concentrations and flowed these samples over captured receptors [23]. Binding of B4/4-GF and BMP-2 to the type I receptor ALK3 was inhibited by Heparin, as indicated by the reduced SPR signal with increasing Heparin concentrations (Fig. 6C). In contrast to the type I receptor ALK3, we found that Heparin stabilized the B4/4-GF and BMP-2 interaction with the type II receptor ActRIIA (Fig. 6D), as indicated by the altered binding kinetics that result mainly in slower off rates. The effect of Heparin on BMPRII binding was more puzzling. The BMPRII–BMP-2 interaction was stabilized by Heparin. However, BMP-4 GF did not bind BMPRII with appreciable affinity and Heparin did not alter its receptor-binding profile (Fig. 6E). Strikingly, Heparin did not significantly affect BMP-4 Δ binding to ALK3 or ActRIIA, as indicated by the near identical SPR signal of all BMP-4 Δ /Heparin concentration ratios. Thus, these results indicate that Heparin differentially affects how ligands interact with their receptors. This effect is likely mediated by N-terminal amino acids that are deleted in BMP-4 Δ . We note that Heparin did not bind the receptors ALK3, ActRIIA, or BMPRII.

To establish how Heparin affects the signaling potency of the different GFs and, thus, gain insights into the biological consequences of Heparin binding, we used a SMAD1/5/8-responsive reporter gene expression assay. We treated cells with a fixed ligand concentration and increasing Heparin concentrations (Fig. 6F). Consistent with our SPR-binding assays, we found that Heparin inhibited BMP-2 and B4/4-GF signaling in a concentration-dependent manner, but not BMP-4 Δ . Taken together, these biophysical and cell-based results indicate that Heparin binds BMP-2 and BMP-4 at their N-terminal sequence to alter how these GFs interact with their receptors and signal.

Discussion

We have established a novel extraction approach to obtain significant quantities of pure and bioactive BMP-4 GF, which can be used for functional and structural studies, as an additive to cell culture media, or as therapeutic for regenerating bone.

BMP-4 is a secreted growth factor that it is expected to accumulate in CM when expressed in mammalian cell culture. However, the purification of BMP-4 GF from CM has been challenging and yields have been impracticably low compared with other secreted mammalian proteins of similar size. Here, we found that the majority of BMP-4 GF is associated with the cell pellet fraction (CPF) of CHO cells, which contains extracellular matrix (ECM) components. This finding is consistent with the previous observation that BMP-4 GF is trapped by the ECM of the cell surface in the *Xenopus* embryo [19]. By contrast, in CM we only detected unprocessed BMP-4 or BMP-4 GF when expressed as a chimeric construct with the BMP-2 pro-domain (B2/4). As we also detected significant amounts of BMP-2 pro-domain in the CM of the B2/4 construct, we speculate that the BMP-2 pro-domain remains associated with the BMP-4 GF to protect the GF moiety from degradation or inhibit its binding to the ECM. We further speculate that, in contrast to the BMP-2 pro-domain, the BMP-4 pro-domain may be unstable after processing, resulting in its degradation (Fig. 7). This model is consistent with the idea that cleavage of a secondary Furin processing site in the BMP-4 pro-domain liberates the mature GF to modulate its localization [29]. Critically, differences in pro-domain stability may help differentiate bioavailability or diffusivity of the related BMP-2 and BMP-4 ligands.

We discovered that BMP-4 GF could be extracted from the CPF/ECM with low ionic strength buffers, detergents, or Heparin. These extraction conditions indicate that both hydrophobic protein–protein and protein-proteoglycan complexes may combine to recruit BMP-4 to the ECM. Thus, in addition to the proteoglycans, other as yet unknown components may recruit BMP-4 to the ECM to restrict BMP-4 diffusion in vivo [19].

We selected one extraction condition that yielded active BMP-4 GF and purified this fraction to homogeneity by adapting previous strategies. Using this approach, we obtained milligram quantities of BMP-4 GF for all constructs, except those expressed as MBP fusions. Although we could purify acceptable amounts of BMP-4 GF from CM when it is expressed as the chimeric B2/4 construct, GF yields from CM were, otherwise, orders of magnitude lower. Thus, the overall benefit of extracting BMP-4 GF from the CPF is an approximately 30- to 100-fold increase in yield over BMP-4 GF purified from CM. Notably, this approach could facilitate GF capture from cultures as working volumes depend on extraction conditions, not culture size. Thus, a CM concentration step is not necessary in this approach.

To ascertain that CPF-purified BMP-4 GF has biochemical and biological activity, we tested binding of the different samples to the receptors ALK3 and ActRIIA, and we assessed their ability to activate SMAD1/5/8 signaling. We found that all CPF/ECM-extracted BMP-4 GF and CM-purified B2/4-GF have similar receptor binding and signaling characteristics, indicating that these fractions are equivalent. Notably, unprocessed BMP-4 did not bind receptors or elicit a SMAD1/5/8 response, indicating that this form is latent. We also benchmarked one representative ECM-purified sample (B4/4 minus furin) with commercial preparations of cell culture made BMP-4 (BMP-4^C), refolded BMP-4 Δ and BMP-2, a GF with nearly identical sequence that is, therefore, considered to have the same biological properties as BMP-4. We found that both cell culture preparations were comparable, whereas BMP-4 Δ and BMP-2 had distinct type II receptor-binding profiles. In addition, BMP-4 Δ had reduced signaling potency. These findings reveal subtle but potentially relevant

functional differences between the different GFs and formulations that may need to be taken into consideration in functional or in vivo studies.

We also analyzed binding of these GFs to Heparin, as this interaction is thought to be critical for BMP-4 ECM association and bioavailability [19]. As expected, both B4/4-GF and BMP-2 bound Heparin with similar association and dissociation rates. However, the affinity of BMP-4 Δ for Heparin was significantly lower, indicating that basic N-terminal sequences in both BMP-4 and BMP-2, which are deleted in BMP-4 Δ , are critical for GF-Heparin binding. We further investigated whether the Heparin-GF interaction affected the molecular and biological activities of BMP-4 and BMP-2. Strikingly, we found that increasing Heparin concentrations suppressed binding of these growth factors to ALK3 and inhibited their signaling. By contrast, Heparin appeared to stabilize binding of these GFs to ActRIIA. As Heparin did not significantly affect the receptor-binding properties or signaling of BMP-4 Δ , we propose that the N-terminal BMP-4 and BMP-2 sequences help regulate BMP-2/4 bioavailability and activity by supporting their interaction with the ECM.

In conclusion, results presented here provide novel insights into the purification, localization, and pro-domain fate of BMP-4 that have implications for its bioprocessing and biological function.

Materials and Methods

Reagents

Human BMP-2 (355-BM) and BMP-4 (314-BP) were obtained from RnD Systems. *E. coli*-derived human BMP-4 Δ (120-05ET) was purchased from PeproTech. Fc-Fusion proteins were expressed and purified in-house.

Expression Plasmids

MBP DNA was obtained from Addgene [30]. Synthetic B4/4 and B2/4 sequences were obtained from GeneArt. Signal peptide and N-terminal His₈ sequences of B4/4 and B2/4 genes were based on the MBP N-terminal sequence. Expression constructs were assembled by PCR and cloned into the pCHO 1.0 vector.

Cell Lines

CHO-S cells were obtained from Thermo Fisher/Gibco and grown at 37 °C, 80% relative humidity, and 8% CO₂ in an orbital shaker. 3×10^7 viable cells in 30 mL culture volume were transfected with 50 μ g linearized vector as per instructions (Thermo Fisher/Gibco). Transfected cells were subjected to static and shake flask selection with 20 μ g/mL puromycin and 200 nM methotrexate until viability reached 85% and cell density exceeded 1×10^6 viable cells/mL. Cell pools were cryopreserved or subjected to a second round of selection with 50 μ g/mL puromycin and 100 nM methotrexate.

Expression

Cryopreserved, stably transfected cell pools were thawed at 37 °C. 30 mL fresh medium was seeded at an approximate density of 3×10^5 viable cells/mL. Fully recovered cell

pools (viability > 90%) were expanded into 300 mL cultures and grown for 10 days with intermittent glucose feeding. Cultures were harvested by centrifugation for 20 min at 14,000 g. Supernatants were filter sterilized. The cell pellet was processed separately.

Immunoblotting

Correspondent volumes of CM, cells, CM + cells, or ECM elution were loaded on 12% SDS–polyacrylamide gels under reducing conditions. Samples were transferred onto a Hybond-P membrane (GE Healthcare). Anti-BMP-4 (Santa Cruz, sc-12721) or Anti-His tag (Takara, 631212) was used as primary antibodies. HRP-conjugated anti-mouse IgG (RnD Systems, HAF018) was used as secondary antibody. WesternBright Sirius HRP substrate was used for detection (Advansta, K-12043-D10). Western blots were visualized by exposing the membrane to autoradiography film.

Purification

BMP-4 GF was captured from CM with a HiTrap Heparin HP affinity column. The column was washed with 20 mM His and 0.5 M NaCl, pH 6.5 and BMP-4 GF was eluted with 20 mM Histidine, 0.1 M NaCl, and 0.6 M Arginine, pH 6.5. BMP-4 was extracted from the CPF by thawing and re-suspending the pellet in 20 mM Histidine, 0.5 M Arginine, and 20% propylene glycol, pH 7.8 (APG). The cell suspension was incubated at 4 °C on a rotating shaker for 1 h and sonicated on ice. Cell debris was removed by ultracentrifugation at 40,000 rpm for 30 min. This sample was diluted tenfold in Butyl Sepharose loading buffer. CM and ECM purification are identical from this point on. BMP-4 GF was loaded onto a HiTrap Butyl HP. The column was washed with 20 mM Tris, 0.4 M NaCl, pH 7.8 and BMP-4 GF was eluted in APG buffer. APG fractions containing BMP-4 were then loaded on inline connected HiTrap Q and HiTrap SP HP ion exchange chromatography columns. The Q column was disconnected and BMP-4 GF was eluted from the SP column in 50 mM KH₂PO₄, 0.5 M Arginine, and 0.4 M NaCl pH 7.0. SP elution fractions were loaded onto a RESOURCE[™] RPC and eluted in a 25% to 45% gradient of 5% acetonitrile, 0.1% TFA (A) and 95% acetonitrile, 0.1% TFA (B). MB4 was captured from CM with a HisTrap excel column and eluted with an imidazole gradient in PBS. The MBP tag was removed with TEV protease and an MBPTrap HP column. Unprocessed BMP-4 was further purified by connecting in line a HiTrap Q HP anion exchange chromatography column and HiTrap SP HP cation exchange chromatography column. The sample was loaded in 50 mM KH₂PO₄, Arginine pH 7.0 and eluted from the SP column with 50 mM KH₂PO₄ and 0.4 M NaCl, pH 7.0.

Surface Plasmon Resonance

Binding affinities and inhibition were determined using a Biacore 3000 SPR instrument. For GF receptor-binding studies, anti-human IgG (Fc) antibody was immobilized onto three channels of a CM5 chip using amine coupling chemistry. 200–300 RU of purified ActRIIA-Fc or ALK3-Fc was captured on the experimental channels. A reference channel was monitored to account for non-specific binding, drift, and bulk shifts. To evaluate ligand fractions, a single sample was injected over captured receptors. To obtain robust ligand-binding affinities, a concentration series of BMP-4 GF was injected over captured receptors. For Heparin inhibition analysis, ligand at a single concentration was incubated with a

concentration series of Heparin. GF-Heparin mixtures were injected over captured receptors. For Heparin-binding analysis, biotinylated Heparin was captured on a streptavidin-coated sensor chip and ligand was injected over captured Heparin. To obtain kinetic binding parameters, data were analyzed with BIAevaluation and Scrubber. To evaluate equilibrium binding, data were analyzed with GraphPad.

Reporter Gene Expression

Approximately 10,000 HepG2 cells per well were seeded in a 96-well plate and grown overnight in DMEM + 10% FBS + 1% pen-strep. Transfection solutions containing 24 µL lipofectamine 2000, assay medium (960 µL growth medium supplemented with 0.1% BSA), 192 ng pGL4.74 [Luc2P/hRluc/TK] vector (control luciferase reporter plasmid, Promega), and 19.2 µg of the SMAD2/3 responsive reporter plasmid pGL4.48 [Luc2P/SBE] or the SMAD1/5/8 responsive reporter plasmid pGL3 [Luc2P/BRE] were prepared and mixed with assay medium after a 30-min incubation. 50 µL of this mixture were added to each well. Transfection medium was removed after 24 h and replaced with assay medium (DMEM + 0.01% BSA) containing test GF and/or Heparin. Cells were incubated for 16 h at 37 °C. Luciferase activity was detected using a dual-glow luciferase assay [31]. Luminescence was measured using a FluoStar Omega plate reader. Relative luciferase units were calculated by dividing firefly luciferase units (FLU) with renilla luciferase units (RLU). Data are expressed as mean of four biological replicates. Error bars correspond to SE of four biological replicates.

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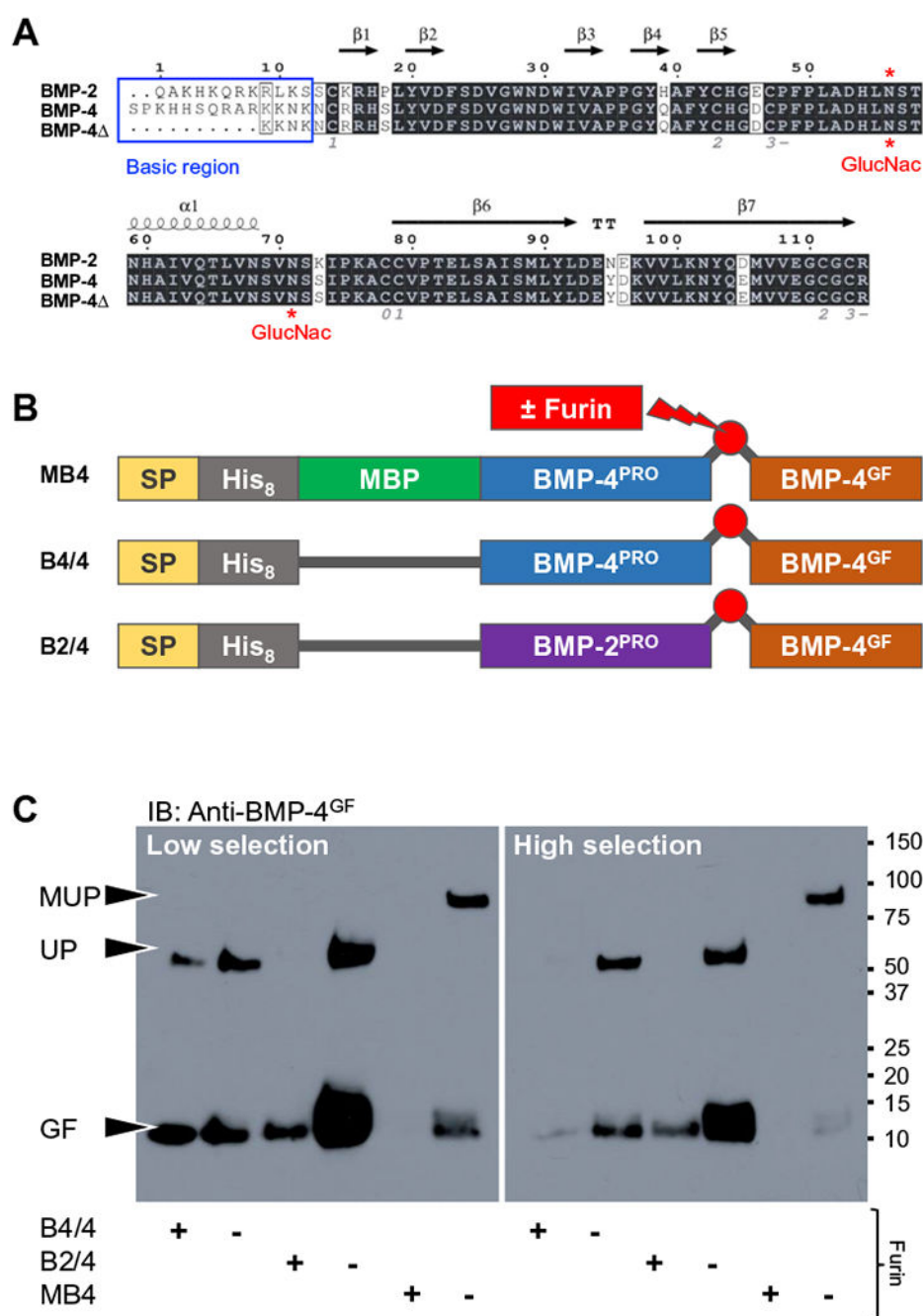
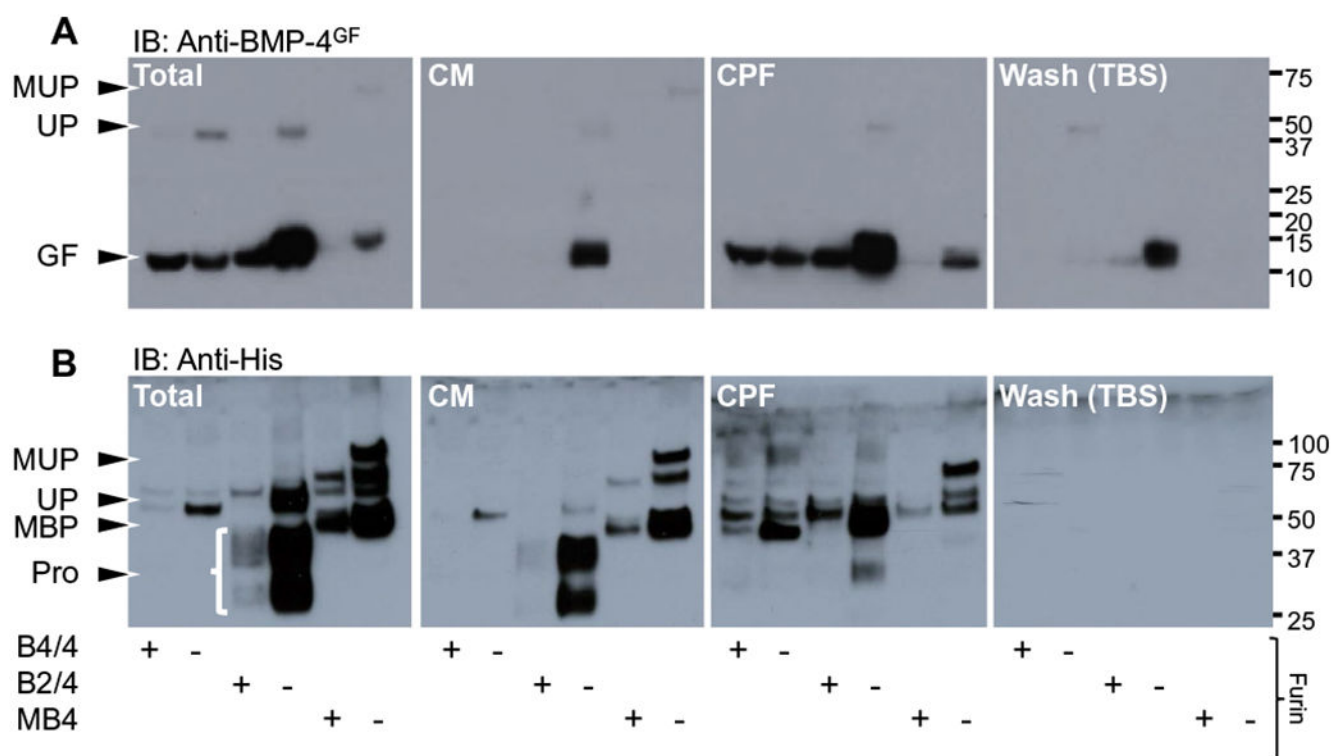
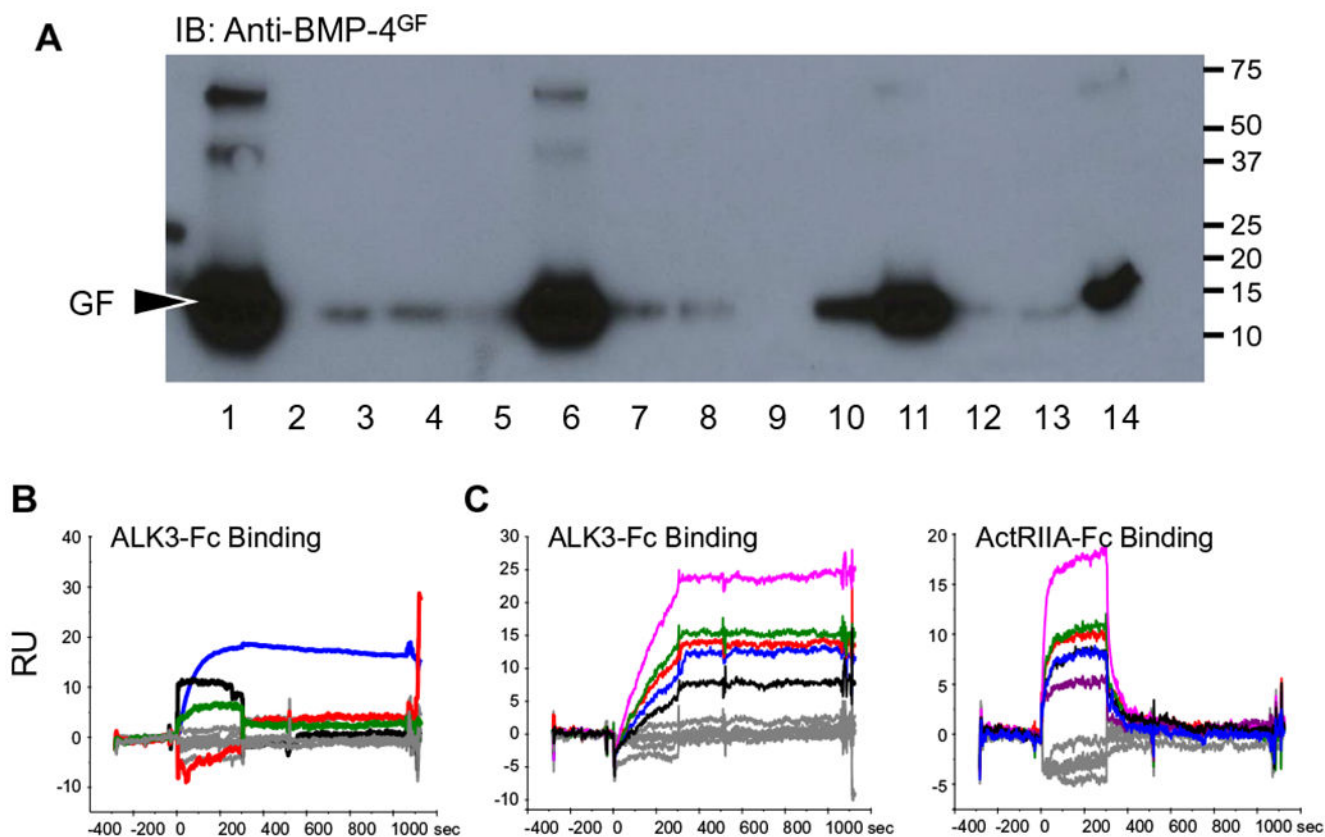


Fig. 1. Construct design. **A** Sequence alignment of BMP-4 and BMP-2 GF region. The basic N-terminal sequence is highly divergent and is deleted in the refolded BMP-4 Δ construct. The main GF sequence following the first Cysteine is highly conserved (92% sequence identity). Identical residues are indicated by the black background. BMP-4 has two N-linked glycans and BMP-2 has one, indicated by the red asterisk. **B** Schematic construct design. A non-native signal peptide followed by a His₈ tag was used in all constructs. One construct included a maltose-binding protein (MBP) sequence that was followed by the

native human BMP-4 sequence (MB4). A second construct consisted of the native human BMP-4 sequence (B4/4). A third construct consisted of the human BMP-2 pro-domain and the BMP-4 GF sequence (B2/4). All constructs were expressed with or without Furin. **C** Western blot using an anti-BMP-4 GF antibody shows BMP-4 levels in CM/cell suspension samples of all constructs $\pm$ furin. The left and right panels show samples subjected to low and high levels of puromycin/methotrexate selection, respectively. The following species are detected: BMP-4 GF (GF) at 10–15 kDa, unprocessed ligand (UP) at 55 kDa and unprocessed MBP fusion (MUP) at 90 kDa. All samples were reduced

**Fig. 2.**

Localization in culture. **A** Western blot using an anti-BMP-4 GF antibody. **B** Western blot using an antibody that detects the His₈ tag N-terminal to the pro-domain. The left panel shows the total sample (CM and cells). The second panel shows cleared CM. The third panel shows the cell pellet. The right panel shows the supernatant of the cell pellet washed with TBS. B4/4, B2/4, and MB4 were expressed $\pm$ furin as indicated. The following species are detected: BMP-4 GF (GF) at 10–15 kDa, unprocessed ligand (UP) at 55 kDa, and unprocessed MBP fusion (MUP) at 90 kDa. All samples were reduced

**Fig. 3.**

Extraction and activity. **A** Supernatant of B4/4 + furin cell pellet fractions (CPF) treated with different buffers. Gel (1) input (CPF), (2) empty, (3) TBS, (4) 1 M NaCl, (5) 2 M NaCl, (6) Na Heparin (100 mg/mL), (7) Na Heparin (200 mg/mL) + 1 M NaCl, (8) 2% Chaps, (9) 2% Tween 20, (10) HNA buffer, (11) APG buffer, (12) 4 mM HCl, (13) 0.5 M HCl, and (14) water. **B** SPR analysis of cell pellet eluted BMP-4. ALK3-Fc was captured on the sensor chip and equivalent volumes of supernatant from cells treated as noted in 3A were injected. APG-eluted BMP-4 (blue curve) shows the most promising binding profile. Other samples did not show appreciable binding activity. **C** SPR analysis of APG extracted BMP-4 from the CPF of different constructs. ALK3-Fc or ActRIIA-Fc were captured on the sensor chip and equivalent volumes of supernatant were injected. APG-extracted CPF supernatants are as follows: blue corresponds to BMP4/4 + fur; red corresponds to BMP4/4 – fur; green corresponds to BMP2/4 + fur; pink corresponds to BMP2/4 – fur; purple corresponds to MBPTevBMP4 + fur; black corresponds to MBPTevBMP4 – fur. Gray lines represent CM samples that lack binding activity

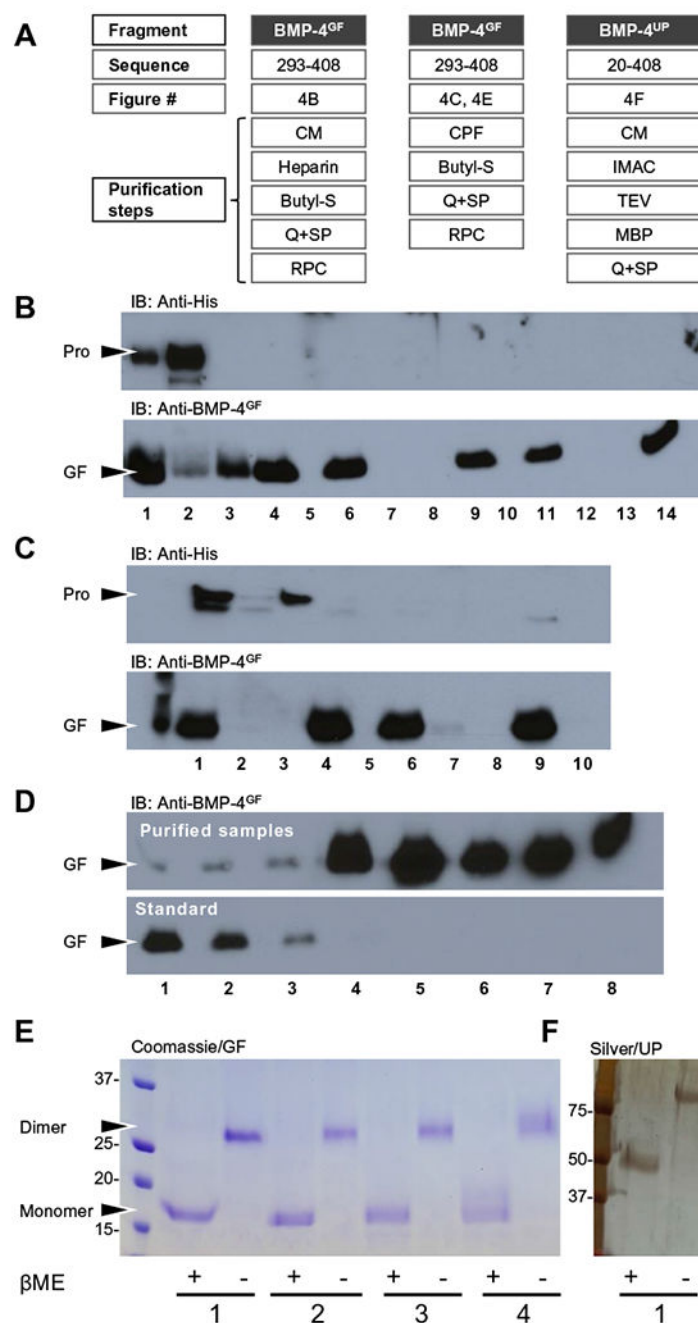
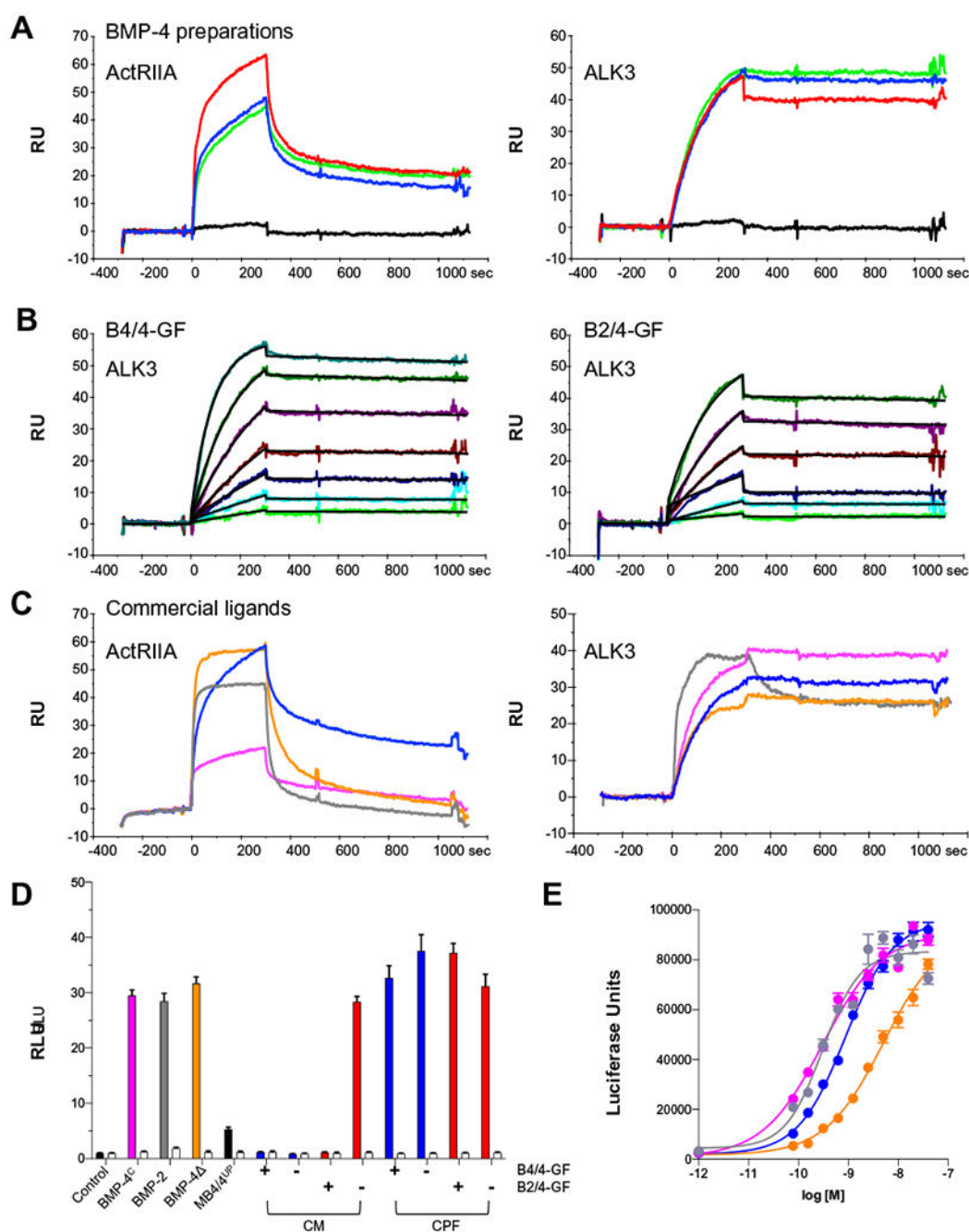


Fig. 4. Purification and quantification. **A** Schematic of purification strategies for MB4 from CM, BMP-4 GF from CM, and BMP-4 GF from cell pellet. **B** Western blot shows purification of BMP-4 GF from CM. In this example, BMP-4 GF was expressed as fusion protein with the BMP-2 pro-domain (B2/4). Samples correspond to (1) CM as Heparin column input, (2) Heparin flow-through (FT), (3) Heparin wash, (4) Heparin elution in HNA buffer, (5) empty, (6) Butyl Sepharose input, (7) Butyl Sepharose FT, (8) Butyl Sepharose wash, (9) Butyl Sepharose elution in APG buffer, (10) empty, (11) Q + SP input, (12) Q + SP FT, (13) Q

elution, and (14) SP elution. The top blot was visualized with an anti-His-tag antibody and shows the pro-domain. The bottom blot was visualized with an anti-BMP-4 GF antibody and shows the GF moiety. **C** Western blot shows purification of BMP-4 GF from the ECM. In this example, BMP-4 GF was expressed with its native pro-domain (B4/4) and co-expressed with Furin. Samples correspond to (1) Butyl Sepharose input, (2) Butyl Sepharose FT, (3) Butyl Sepharose wash, (4) Butyl Sepharose elution in APG buffer, (5) empty, (6) Q + SP input, (7) Q + SP FT, (8) Q elution, (9) SP elution, and (10) SP regeneration. The top blot was visualized with an anti-His-tag antibody and shows the pro-domain. The bottom blot was visualized with an anti-BMP-4 GF antibody and shows the GF moiety. **D** Western blot quantification. The top blot shows equivalent samples of purified BMP-4 corresponding to (1) CM B4/4 + Furin, (2) CM B4/4 – Furin, (3) CM B2/4 + Furin, (4) CM B2/4 – Furin, (5) CPF B4/4 + Furin, (6) CPF B4/4 – Furin, (7) CPF B2/4 + Furin, and (8) CPF B2/4 – Furin. The bottom blot shows commercial BMP-4 (RnD). Loaded sample amounts are (1) 8 ng, (2) 4 ng, (3) 2 ng, and (4) 1 ng. The blots were visualized with an anti-BMP-4 GF antibody and shows the GF moiety, top, and bottom blots were exposed on the same film. **E** Coomassie-stained gel of RPC purified BMP-4^{GF} loaded under reducing (monomer, + β ME) and non-reducing (dimer, – β ME) conditions. The blot shows equivalent samples of purified BMP-4 corresponding to (1) CPF B4/4 + Furin, (2) CPF B4/4 – Furin, (3) CPF B2/4 + Furin, and (4) CPF B2/4 – Furin. Molecular weights in kDa are noted next to the marker. **F** Silver- stained gel of purified MB4 expressed without furin. The gel was loaded under reducing (monomer, + β ME) and non-reducing (dimer, – β ME) conditions. The MBP moiety was removed by TEV cleavage. Only unprocessed BMP-4 is observed. Molecular weights in kDa are noted next to the marker

**Fig. 5.**

Receptor binding and signaling. **A** Surface plasmon resonance (SPR) comparison of different BMP-4 samples. The left panel shows binding to the receptor ActRIIA, and the right panel shows binding to the receptor ALK3. 40 nM BMP-4 GF as determined by BCA was injected over the sensor chip coated with captured receptors. The GF samples were obtained as follows. Green: GF from B2/4 CM; red: from B2/4 CPF; blue: GF from B4/4 CPF; black: unprocessed BMP-4 from MB4 CM. **B** SPR analysis of ALK3-BMP-4 binding. GF samples were obtained as follows. Left panel: GF from B4/4 CPF; right panel: GF from

B2/4 CPF. GF concentrations were as follows: teal 80 nM, green 40 nM, purple 20 nM, brown 10 nM, blue 5 nM, cyan 2.5 nM, and light green 1.25 nM. **C** SPR comparison with commercial samples and BMP-2. The left panel shows binding to the receptor ActRIIA, and the right panel shows binding to the receptor ALK3. 40 nM BMP-4 GF was injected over the sensor chip. The GF samples were obtained as follows. Blue: GF from B4/4 CPF; magenta: RnD Systems; orange: BMP-4Δ; gray: RnD Systems BMP-2. **D** Comparison of SMAD1/5/8 signaling. HEK293 cells stably transfected with a SMAD1/5/8 responsive luciferase reporter were subjected to 10 nM GF as indicated. The white bar represents the corresponding SMAD2/3 signaling readout. **E** Comparison of signaling potency. HepG2 cells transiently transfected with a SMAD1/5/8 responsive luciferase reporter were subjected to increasing GF concentrations as indicated. Color coding of GF samples is the same as above

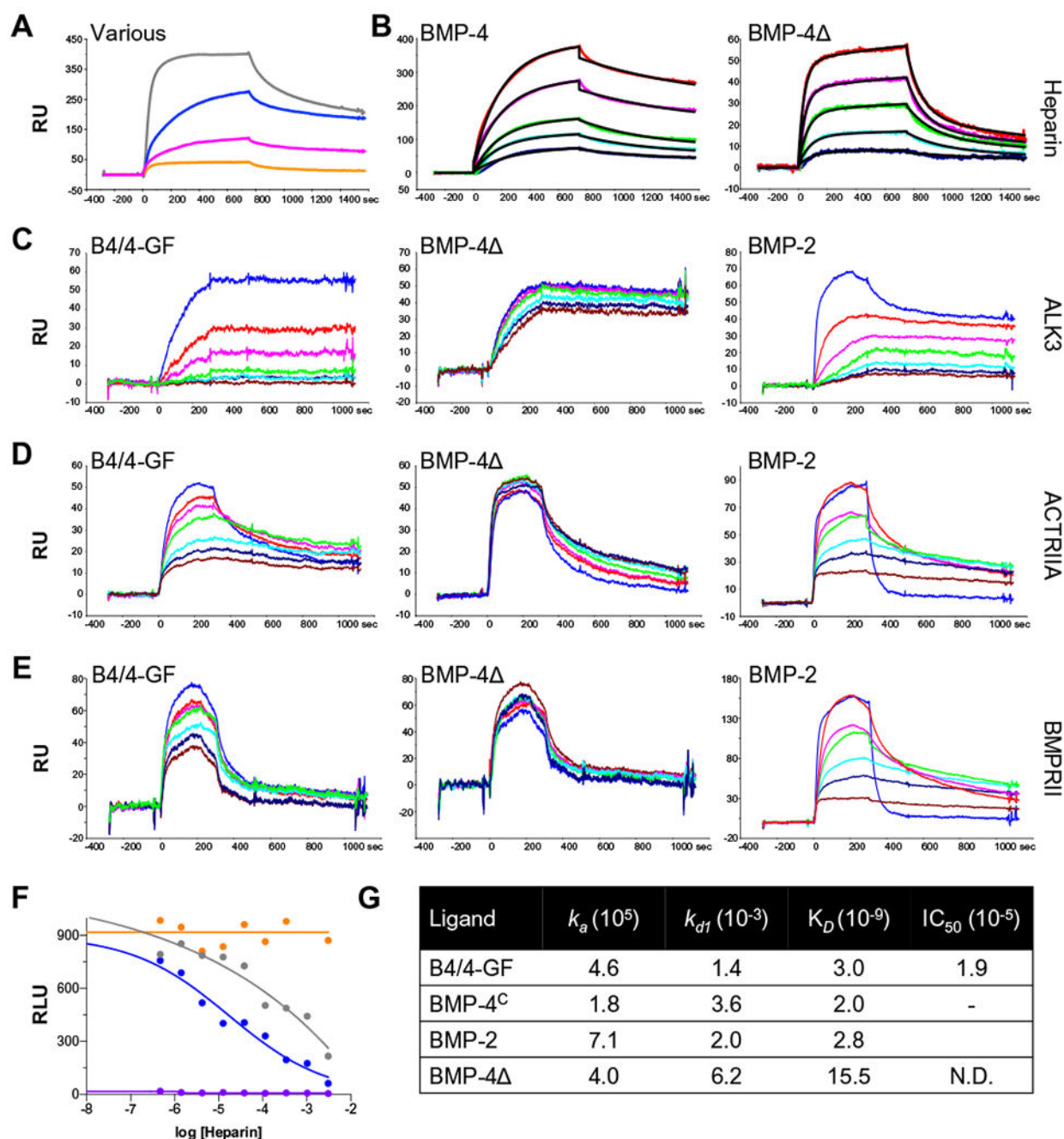


Fig. 6. Heparin interaction. **A** SPR comparison of BMP-4 binding to Heparin. 25 M GF was injected over a sensor chip coated with captured receptors. GF samples are as indicated in Fig. 5. **B** SPR titration analysis of Heparin–BMP-4. The left panel shows binding of GF obtained from the B4/4 construct CPF. The right panel shows binding of BMP-4Δ. GF concentrations are as follows. Light red: 100 nM; magenta: 50 nM; light green: 25 nM; cyan: 12.5 nM; blue: 6.25 nM; red: 3.125 nM. **C** SPR analysis of Heparin effect on the BMP-4–ALK3 interaction. **D** SPR analysis of Heparin effect on the BMP-4–

ActRIIA interaction. The left panel shows binding of GF obtained from the B4/4 CPF, the middle panel shows BMP-4 Δ , and the right panel shows BMP-2. 40 nM GF preincubated with increasing Heparin concentrations was injected over captured receptors. Heparin concentrations are as follows: blue: no Heparin; red: 0.123 μ M; magenta: 0.370 μ M; light green: 1.1 μ M; cyan: 3.3 μ M; dark blue: 10 μ M; dark red: 30 μ M. **E** SPR analysis of Heparin effect on the BMP-4–BMPRII interaction. The left panel shows binding of GF obtained from the B4/4 CPF, the middle panel shows BMP-4 Δ , and the right panel shows BMP-2. 40 nM GF preincubated with increasing Heparin concentrations was injected over captured receptors. Heparin concentrations are as follows: blue: no Heparin; red: 0.123 μ M; magenta: 0.370 μ M; light green: 1.1 μ M; cyan: 3.3 μ M; dark blue: 10 μ M; dark red: 30 μ M. **F** Comparison of inhibitory potency. HepG2 cells transiently transfected with a SMAD1/5/8 responsive luciferase reporter were subjected to 40 nM GF preincubated with increasing Heparin concentrations. Color coding of GF samples is the same as in Fig. 5. In addition, the violet line represents Heparin control. **G** Heparin binding and inhibition analysis. k_{on} , k_{off} and K_d values were obtained from SPR titration analyses. IC₅₀ values were obtained from fitting the dose-response curves in 6E

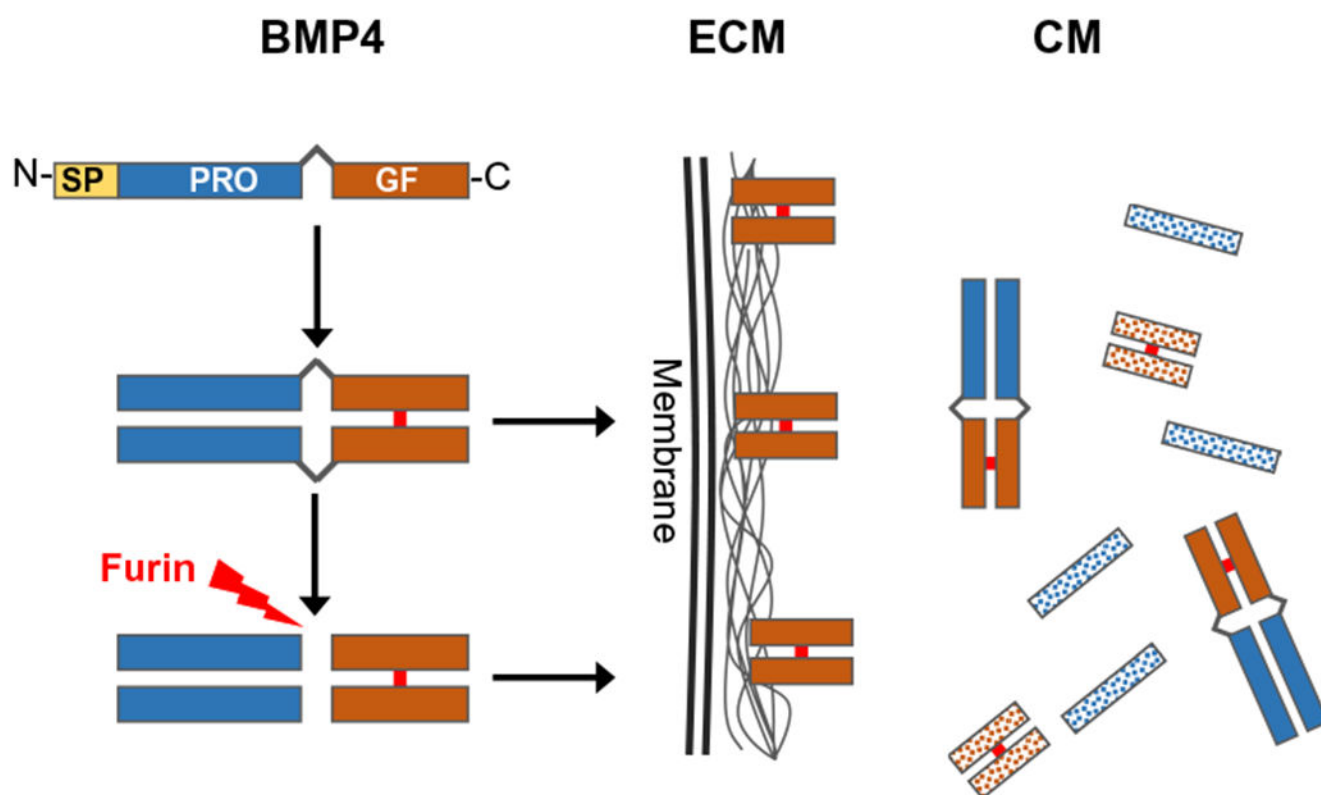


Fig. 7.

Model of GF localization. BMP-4 is cleaved by Furin family proteases during the process of biosynthesis and secretion to release the mature GF. Unprocessed, secreted BMP4 is mainly found in the CM. By contrast, the mature BMP-4 GF domain is mainly found associated with cells or the cell membrane or ECM. This interaction may be required for ligand stability as GF levels in the CM are negligible. The BMP-4 pro-domain may not remain associated with the GF domain or may be degraded after processing, as it does not copurify with the GF and only low levels are detected in the ECM or CM. BMP-4 GF is only found in the CM when expressed as a fusion protein with the BMP-2 pro-domain, indicating that the BMP-2 pro-domain may remain associated with the GF moiety and may be resistant to proteolysis

Table 1

Estimated yields of purified BMP-4

Construct	Furin	Fraction	Purified moiety	Final purification	Yield (per L) ^a
B4/4	+	CM	GF	RPC	< 20 µg
B4/4	-	CM	GF	RPC	< 20 µg
B2/4	+	CM	GF	RPC	< 20 µg
B2/4	-	CM	GF	RPC	~ 100 µg
B4/4	+	CPF	GF	RPC	~ 650 µg
B4/4	-	CPF	GF	RPC	~ 650 µg
B2/4	+	CPF	GF	RPC	~1000 µg
B2/4	-	CPF	GF	RPC	~1900 µg
MB4	+	CM	Degraded	N/A	N/A
MB4	-	CM	FL	Q + SP	~ 400 µg

^a 300 mL cultures seeded at 3×10^5 cells/mL were harvested after 10 days of growth. Volumes were matched throughout purification. Yields were estimated by BCA

Table 2

Receptor binding and signaling analysis

	ActRIIA		ALK3		Signaling		
	k_a (10^5)	k_d (10^{-3})	K_D (10^{-9})	k_a (10^5)	k_d (10^{-5})	K_D (10^{-11})	EC_{50} (10^{-10})
B4/4-GF	3.3	1.2	3.5	5.1	1.4	2.8	8.7
BMP-4	1.2	23.0	1.9	5.1	2.5	4.9	3.0
BMP-4Δ	8.3	6.3	7.2	5.9	1.3	2.2	0.5
BMP-2	8.0	9.8	0.1	6.4	0.4	0.7	3.0